

Decent Coaching Classes
Std. VIII (CBSE) — Mind Your Math - I
Answer Key — Squares and Square Roots

Section A

- (1) (d) 1156 [$34^2 = 1156$]
- (2) (a) 9 [$87^2 = 7569$]
- (3) (a) 18 [numbers between n^2 and $(n+1)^2 = 2n$; here $n = 9$]
- (4) (b) (6, 8, 10) [$6^2 + 8^2 = 36 + 64 = 100 = 10^2$]
- (5) (c) 15

Section B

- (1) Sum of first 7 odd numbers = $7^2 = 49$
- (2) $35^2 = (30 + 5)^2 = 30^2 + 2 \times 30 \times 5 + 5^2 = 900 + 300 + 25 = 1225$
- (3) 15625 has 5 digits (odd); number of digits in square root = $(5+1)/2 = 3$ [$\sqrt{15625} = 125$]
- (4) $2352 = 2^4 \times 3 \times 7^2$. The factor 3 is unpaired, so 2352 is not a perfect square. Multiply by 3 $\rightarrow 7056 = 84^2$, a perfect square. Smallest number required = 3.
- (5) $\sqrt{0.0081} = \sqrt{81/10000} = 9/100 = 0.09$

Section C

- (1) $4096 = 2^{12}$. $\sqrt{4096} = 2^6 = 64$
- (2) By long division method, $\sqrt{1764} = 42$
- (3) $44^2 = 1936$ and $45^2 = 2025$. Since 1989 lies between these, subtract $1989 - 1936 = 53$. Resulting number = 1936, and $\sqrt{1936} = 44$.

Section D

- (1)(a) $80^2 = 6400$ and $81^2 = 6561$. Since 6412 lies between these, add $6561 - 6412 = 149$. Resulting number = 6561, and $\sqrt{6561} = 81$.
- (1)(b) By long division method, $\sqrt{51.84} = 7.2$